

SAI KRISHNA KAUSHIK PINNELLI

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PROFESSIONAL SUMMARY

Data-driven analytics professional with 8+ years of experience analyzing large-scale behavioral data, designing A/B tests, and measuring system performance across e-commerce, financial services, and insurance domains. Strong expertise in SQL, Python (pandas, matplotlib), and Spark for data manipulation and statistical analysis. Proven experience designing and interpreting statistical tests (A/B tests, hypothesis testing, confidence intervals) to measure business impact and validate model performance. Adept at partnering with data scientists, ML engineers, and cross-functional teams to translate complex findings into clear narratives that drive product decisions and operational efficiency. Expert in building dashboards using Power BI and Tableau to track key metrics and customer preferences over time.

EDUCATION

George Mason University	Fairfax, VA
Master of Science in Data Analytics Engineering	May 2023
Gandhi Institute of Technology and Management	Hyderabad, India
Bachelor of Technology in Electrical and Electronics Engineering	May 2016

EXPERIENCE

Oregon Department of Human Services, Salem, OR	Aug 2025 – Present
Senior Data Analyst	

- Led enterprise analytics initiatives supporting 10+ programs, partnering with cross-functional teams in Product, Strategy, and Technology to translate complex business requirements into scalable reporting and BI solutions, improving data-driven decision-making.
- Designed and developed interactive Power BI dashboards using advanced DAX measures and calculated columns to define KPIs, performance metrics, and quarterly target review (QTR) indicators.
- Utilized Power Query for data transformation, cleansing, and integration across multiple structured data sources, standardizing reporting workflows and improving data consistency.
- Built advanced Excel-based reporting models using Pivot Tables, Power Query, complex formulas, and structured validation controls to automate recurring reports and reduce manual effort by 30%.
- Developed and maintained 20+ integrated SQL datasets, ensuring data accuracy, completeness, and standardized business definitions across enterprise reporting platforms.
- Implemented self-service analytics frameworks within Power BI, enabling program managers and stakeholders to independently access dashboards, conduct drill-down analysis, and generate ad-hoc insights.
- Established data governance documentation including metric definitions, transformation logic, metadata tracking, and lineage mapping to support audit readiness and reporting transparency.
- Built automated data quality validation checks and reconciliation processes to monitor reporting accuracy and minimize data discrepancies.
- Conducted trend analysis and benchmarking studies using Power BI and Excel to identify operational performance gaps and support strategic improvement initiatives.
- Collaborated with cross-functional stakeholders and IT teams to manage secure data access controls and ensure compliance with internal data governance policies.
- Led and mentored a team of 3 analysts, overseeing analytics delivery, quality assurance, and stakeholder communication.

Smart IT Pros, Inc, Rockwall, TX	July 2023 – Aug 2025
Business Data Analyst	

- Analyzed large-scale behavioral data (2M+ interactions) to identify customer trends and conducted A/B tests, improving engagement by 20%.
- Designed and optimized SQL data models and built scalable ETL pipelines to integrate multi-source data with high quality and consistency.
- Developed Power BI dashboards tracking KPIs (CTR, conversion, retention), improving efficiency by 50% and forecasting accuracy by 30%.
- Implemented data quality checks, validation frameworks, and anomaly detection to ensure reliable datasets.
- Translated customer feedback and behavioral insights into product improvements, increasing retention by 15%.
- Automated reporting workflows and standardized data processes to enable scalable self-service analytics.
- Built KPI scorecards and performance monitoring frameworks to support business growth and decision-making.
- Collaborated with product, engineering, and business teams to deliver actionable analytics solutions.
- Monitored system performance and customer trends to identify optimization opportunities.
- Delivered Power BI training and documentation, increasing analytics adoption by 40%.

Business Data Analyst - Internship

- Performed data cleaning, transformation, and validation using Excel and PostgreSQL to improve data accuracy and reduce processing time by 30%.
- Developed advanced analytics solutions using Python, Power BI, and PostgreSQL, including predictive modeling and interactive dashboards, improving project efficiency by 15%.
- Built and optimized structured datasets to support KPI tracking, trend analysis, and operational performance monitoring.
- Applied statistical analysis and exploratory data analysis (EDA) to identify key performance drivers and business improvement opportunities.
- Generated actionable insights that increased revenue by 15% and reduced operational expenses by 10%.
- Presented analytical findings, dashboards, and data-driven recommendations to stakeholders to support strategic and operational decision-making.

Tech Mahindra Limited, Hyderabad, India

Oct 2016 – Aug 2021

Data Engineer

- Led large-scale data engineering initiatives for CitiBank e-commerce credit card portfolios, processing high-volume transactional and behavioral datasets (terabytes of customer interaction data) to support risk analytics, model validation, and customer performance monitoring.
- Designed, developed, and optimized scalable ETL pipelines using Apache Spark, Python, SQL, Hadoop, SSIS, processing terabytes of structured and transactional data.
- Built integrated batch and real-time data workflows, reducing pipeline latency by 30% and accelerating operational performance by 25%.
- Implemented Snowflake as a centralized cloud data warehouse, improving query performance, storage optimization, and long-term scalability.
- Orchestrated complex data workflows using Apache Airflow across AWS and on-prem environments to ensure reliable and automated processing.
- Developed automated data quality validation checks and error-handling frameworks, minimizing data anomalies and strengthening compliance controls.
- Supported AWS-based cloud migration initiatives, enabling scalable compute and storage architecture.
- Created executive-level Tableau dashboards and SSRS reports aligned to financial and operational KPIs, improving reporting efficiency and increasing stakeholder satisfaction by 15%.
- Automated reporting and operational tasks using MS SQL Server and Excel macros, reducing manual effort by 30%.
- Created detailed Source-to-Target Mapping (STTM) documents and documented transformation rules to support data migration and audit validation processes.
- Performed manual data profiling to assess data quality, completeness, and integrity prior to transformation and loading.
- Partnered with cross-functional IT, BI, and business teams to gather requirements, conduct UAT, and deliver production-ready data solutions.
- Contributed to GDPR-aligned audit validation cycles and ensured regulatory compliance across transformation pipelines.

PROJECTS**Data Analysis on Identifying Non-Traditional Cyber Security Candidates (Tools: Python, SQL, NLP, AWS EC2, Tableau)**

- Built an NLP pipeline (TF-IDF, NER, LDA topic modeling) to analyze 10K+ cybersecurity job postings and extract skill patterns.
- Processed and stored structured/unstructured data using Python and optimized SQL queries on AWS EC2.
- Developed Tableau dashboards to visualize diversity gaps and alternative career pathways, improving insight generation by 30%.

Data Analysis on Accidents in California (Tools: Python, Pandas, Altair, Statistical Analysis)

- Cleaned and transformed large accident datasets using Pandas (feature engineering, aggregations, missing value handling).
- Applied statistical methods (correlation analysis, hypothesis testing, regression) to identify key accident risk factors.
- Built interactive Altair visualizations highlighting weather impact, time-based risk, and geographic hotspots.

Machine Learning Project on Credit Card Account Data (Tools: R, tidymodels, ggplot2, tidyverse, Statistical Modeling)

- Developed and tuned classification models (Logistic Regression, Random Forest) using tidymodels for churn prediction.
- Engineered behavioral features (utilization ratios, transaction frequency, delinquency trends) to improve model accuracy by 20%.
- Evaluated performance using ROC-AUC and lift analysis, translating outputs into customer retention strategies.

TECHNICAL SKILLS

- **PROGRAMMING LANGUAGES:** SQL, R, Python, SnowSQL, Oracle SQL
- **TECHNOLOGIES:** AWS (EC2, S3), RDBMS, GitHub, Microsoft Office, RStudio, Agile, Jira, Scrum, Google Collab, Jupyter Notebook
- **DATA ANALYTICS TOOLS:** Power BI (DAX), Excel Models, Tableau, MySQL, MS SQL Server, PostgreSQL, Snowflake, Pandas, NumPy
- **STATISTICAL METHODS:** Regression analysis, Hypothesis testing, Classification, Clustering, Decision Trees, A/B Testing
- **BUSINESS INTELLIGENCE:** Data modeling, Data mining, Data entry, Data visualization, Forecasting
- **CERTIFICATIONS:** PCEP-30-01 (Certified Python Programmer)